Gene Drug Visualizer User Manual

v1.0.0

1 Introduction

1.1 Purpose

Our software is designed to build an intuitive and efficient drug-gene association analysis platform. By integrating transcriptomic data and drug sensitivity data from a large number of cell lines, it helps users rapidly explore potential associations between gene expression levels and drug response. Based on the gene name selected by the user, the software automatically analyzes differences in drug sensitivity between the high- and low-expression groups for that gene and visually presents statistically significant results through various interactive visualization methods (such as violin plots and Z-score plots).

1.2 References

Pandas library: A powerful open-source data structure and tool library for efficient and convenient data analysis and manipulation in Python, used as a reference for the software's core data-processing module.

Matplotlib library: The most widely used and fundamental standard library for visualization and plotting in the Python ecosystem, used as a reference for the drawing logic and graphical element settings of the software's basic data-visualization functions.

SciPy library: A core open-source software library built on Python and NumPy for mathematical, scientific, and engineering computing, used as a reference for implementing statistical significance calculations and numerical computing functions in the software.

GDSC database: A large resource database linking cancer drug sensitivity with genomic features, used as a reference for gene identifier mapping and annotation, as well as for defining gene-expression quantification standards in the software.

1.3 Terms and Abbreviations

IC_{50}: Half-maximal inhibitory concentration, a key indicator for measuring drug sensitivity.

AUC: Area under the curve, a parameter used to evaluate the overall intensity of a drug's effect.

Z-score: A standardized value used to evaluate how far a data point deviates from the overall mean and to help identify outliers.

p-value: An indicator of statistical significance. Following generally accepted standards, significance levels are marked with asterisks (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

2 Software Overview

2.1 Software Functions

2.1.1 Data Query and Filtering

Supports entering a gene name to query the drug sensitivity associated with high and low expression of that gene across multiple cell lines.

Provides quantile options for grouping high and low gene expression (such as 2-quantile and 3-quantile grouping). The default is 2-quantile grouping, and users can customize the grouping method.

2.1.2 Statistical Analysis

Automatically calculates significant differences (p-values) between the high- and low-expression groups for drug-sensitivity indicators (IC50 and AUC) and marks them with asterisks ($p < 0.05$, $p < 0.01$, $p < 0.001$).

Calculates Z-scores to help determine whether differences in drug sensitivity are driven by outliers.

2.1.3 Interactive Visualization

Main view: Displays the association between gene expression and drug sensitivity as a scatter plot. Users can click data points to view detailed information.

Gene-expression violin plot: Displays the distribution of gene expression in different tissues or tumor types, with expression values on the vertical axis and groups on the horizontal axis.

Scatter plot: Displays the correlation between gene-expression level and drug sensitivity, with drug sensitivity on the vertical axis and expression value on the horizontal axis, and labels the correlation R value.

2.1.4 Data Details Display

Clicking a data point displays drug information (such as the name and mechanism of action), gene-expression level (such as the FPKM value), and the expression-drug sensitivity correlation, enabling users to gain a deeper understanding. Data export is also supported for further analysis.

2.1.5 Data Export and Sharing

Supports chart export in PNG/SVG format and PDF export for use in papers or reports.

2.2 Software Operating Environment

Minimum configuration:

Processor: 1 GHz or higher

Memory: 4 GB RAM

Storage space: 1 GB of available space

Recommended configuration:

Processor: 2 GHz dual-core or higher

Memory: 8 GB RAM or more

Storage space: 4 GB of available space

2.3 System Requirements

2.3.1 Operating System

Windows: Windows 10/11 (64-bit)

2.3.2 Runtime Dependencies

No additional dependencies are required. The environment dependencies needed for operation have already been integrated into the compressed package.

3 Using the Software

3.1 Software Initialization

After extraction, click gene-drug-app.exe to open the software.

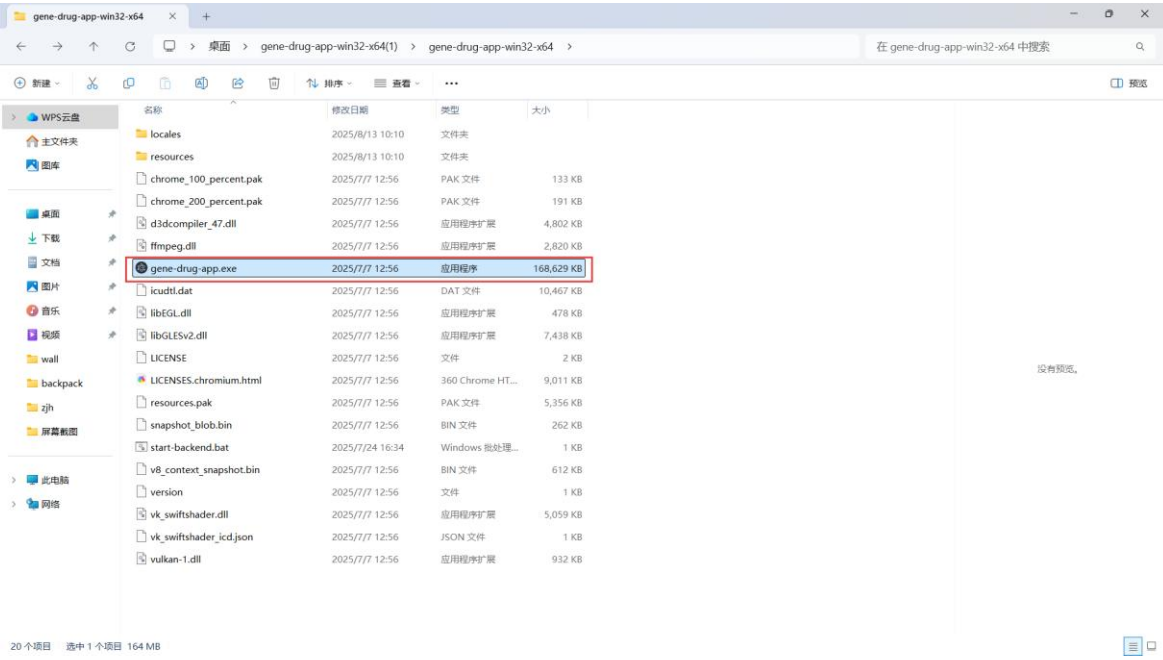

Figure 3.1 Software Login

3.2 Main Interface Overview

As shown in the figure, the platform's main interface consists of a top title bar, an upper control panel, and a lower visualization display area. The upper control panel contains: a gene search bar; a site drop-down box (including 29 sites such as kidney, soft tissue, ovary, bone, central nervous system, upper aerodigestive tract, stomach, prostate, esophagus, autonomic ganglia, pleura, skin, testis, placenta, vulva, urinary tract, thyroid, small intestine, lung, liver, large intestine, and others); a histology grouping drop-down box (including 30 histological types such as neuroblastoma, nephroblastoma, synovial sarcoma, sarcoma, rhabdomyosarcoma, rhabdoid tumor, medulloblastoma, osteosarcoma, mesothelioma, malignant melanoma, liposarcoma, lymphoid neoplasm, leiomyosarcoma, leiomyoblastoma, hyperplasia, hematopoietic system tumor, glioma, adrenocortical tumor, and others); a TCGA cancer-type filter-label drop-down box (classifying cell lines according to TCGA and including 32 labels such as MB, HNSC, PRAD, ESCA, LIHC, DLBC, NB, MESO, SKCM, MM, LAML, CLL, ALL, LCML, LGG, GBM, CESC, BLCA, STAD, and OTHER); an indicator-selection drop-down box (including four indicators: Z-score, natural logarithm of IC50, area under the curve, and root mean square error); and grouping-method switching buttons (including five grouping methods: median, upper tertile, lower tertile, upper quartile, and lower quartile).

The lower display area is blank by default and dynamically displays charts and data tables after the user selects parameters.

After entering the software interface, drug-sensitivity information for the TP53 gene is displayed by default, using Z-score as the indicator and the median as the grouping method.

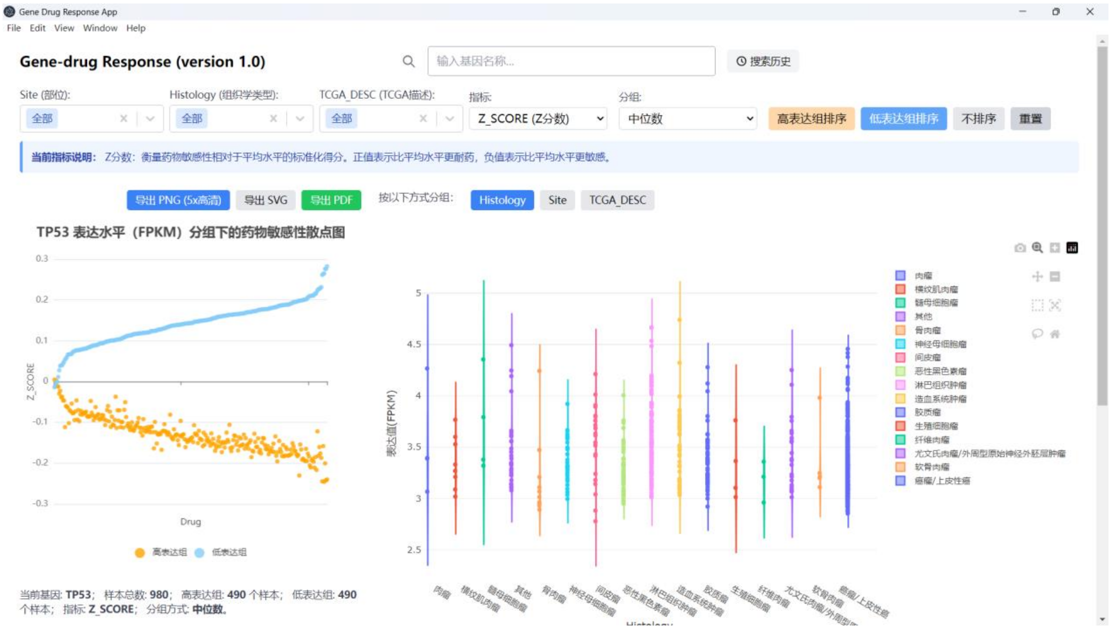

Figure 3.2 Main Interface Overview

3.3 Gene Search Bar

A partial string of a gene name can be entered for searching. As the user types, a list of matching genes automatically appears for selection. After a gene is selected, the system begins loading data for that gene (for example, BCL2).

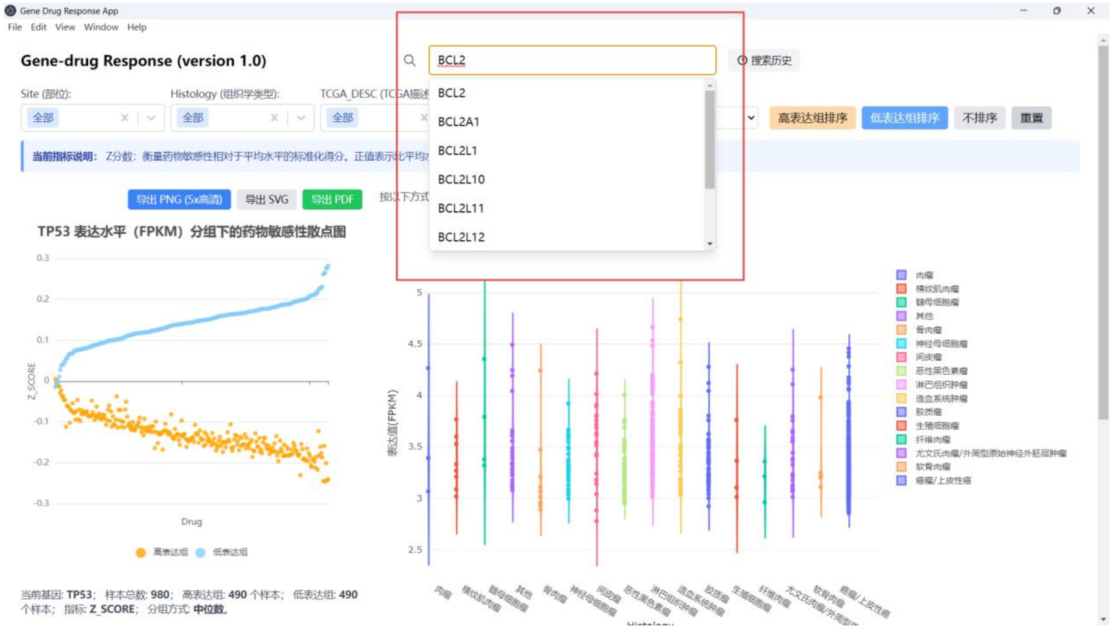

Figure 3.3 Gene Search Bar Pop-up Window

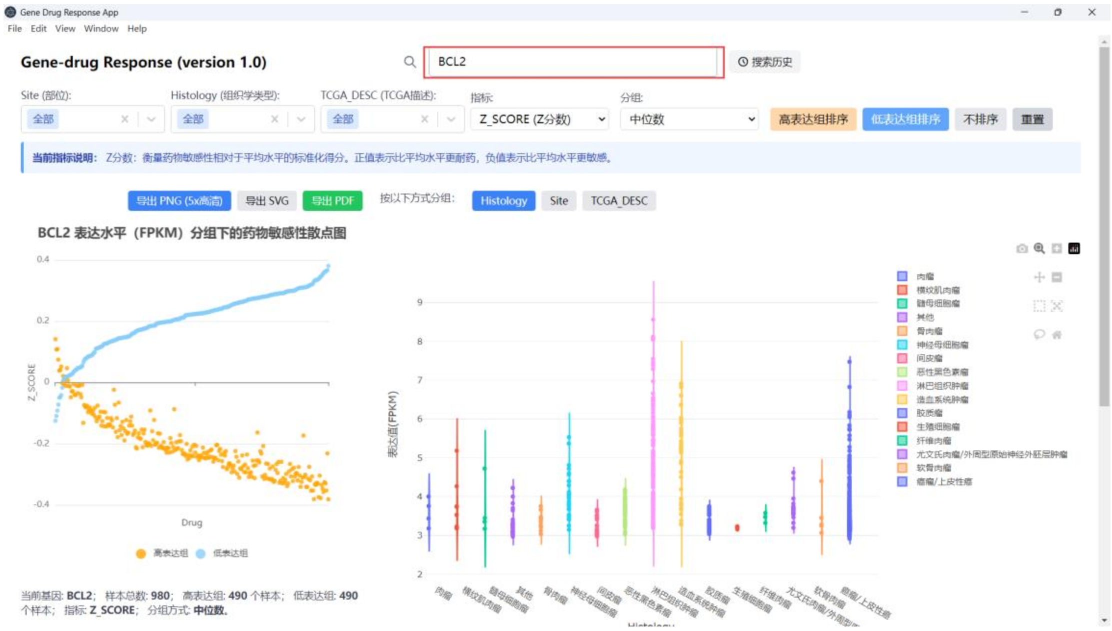

Figure 3.4 Gene Search Bar

3.4 Histology and Other Type Filtering

The drop-down menu lists the histological types included in the dataset (such as lung cancer and breast cancer). Users can select a specific type to analyze only that subset; the default value, "All," means that no filtering is applied and all cell-line data are used.

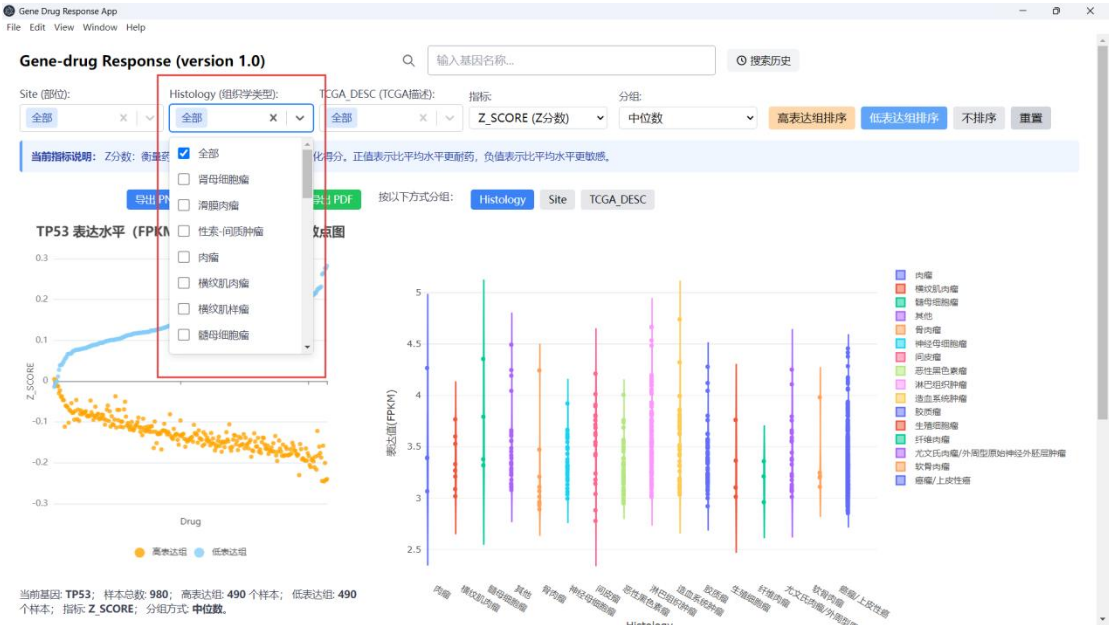

Figure 3.5 Histological Type Selection Bar

3.5 TCGA Label Filtering

Provides TCGA codes for tumor classifications (such as LUAD for lung adenocarcinoma and BRCA for breast cancer). This function is similar to tissue filtering and can filter data according to more standardized cancer-type classifications. Generally, one of the two is used (if a specific tissue is selected, the TCGA label is restricted accordingly).

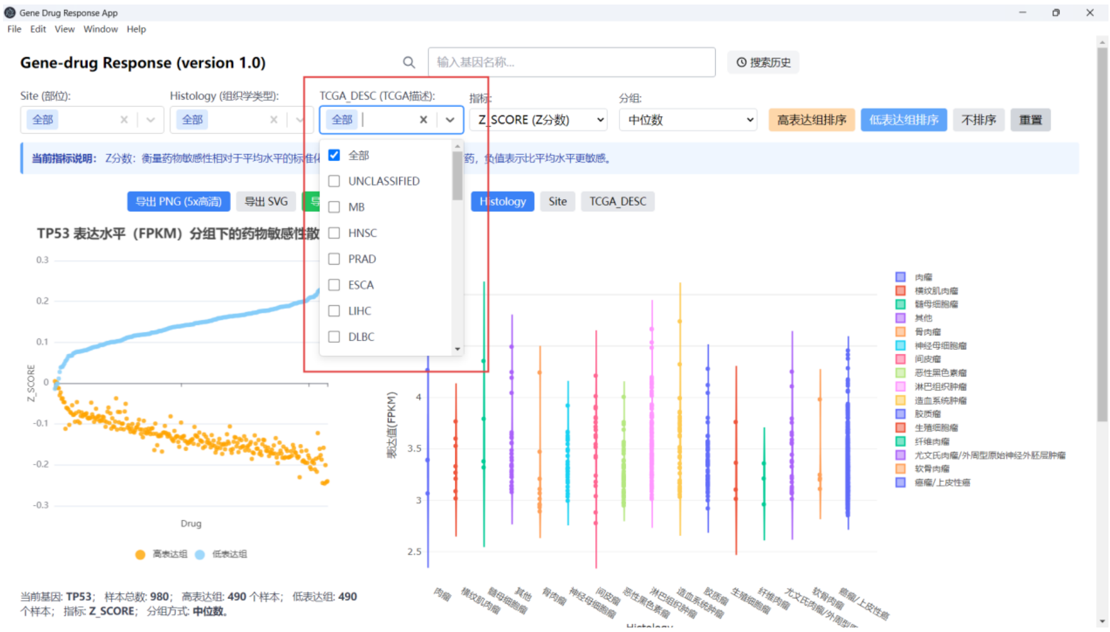

Figure 3.6 TCGGA Label Selection Bar

3.6 Indicator Selection

Users can select one of four drug-sensitivity indicators of interest: Z-score, LN(IC50), AUC, and RMSE (the default indicator is Z-score). The meanings of the indicators differ slightly. Z-score: a standardized score measuring drug sensitivity relative to the average level; a positive value indicates greater resistance than average, while a negative value indicates greater sensitivity than average. LN_IC50: the natural logarithm of the IC50 value; IC50 is the drug concentration required to inhibit 50% of cell growth, and a smaller value indicates a better drug effect. AUC: the area under the dose-response curve, reflecting the overall inhibitory effect of the drug at different concentrations; a larger value indicates a stronger drug effect. RMSE: root mean square error, used to measure the accuracy of the drug-response prediction model; a smaller value indicates a more accurate prediction. Selecting an indicator changes the type of values presented in the drug-response plot.

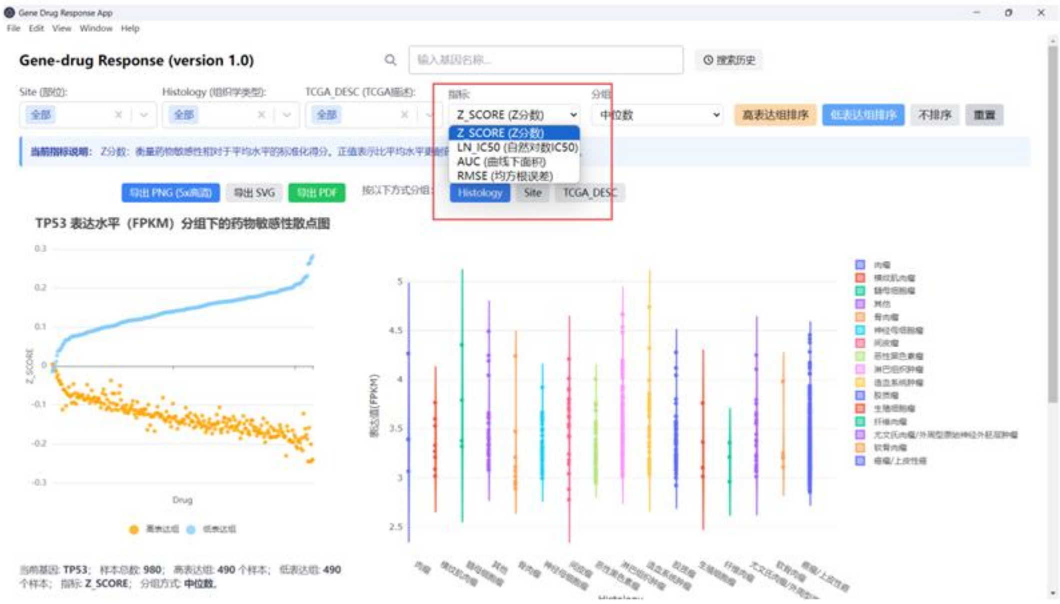

Figure 3.7 Indicator Selection Bar

3.7 Grouping Method

Several buttons are provided to quickly set the criteria for dividing high and low gene expression, such as "median grouping" and "quartile grouping." Median grouping calculates the median expression of the corresponding gene across all samples and uses the median as the threshold. The high-expression group contains samples above the median, and the low-expression group contains samples at or below the median. Tertile grouping is performed according to tertiles (33% and 67%): the lower tertile contains the lowest one-third of samples, and the upper tertile contains the highest one-third of samples. Quartile grouping divides expression values according to quartiles (Q1 = 25%, Q2 = median, Q3 = 75%): the lower quartile contains the 25% of samples with the lowest expression, and the upper quartile contains the 25% of samples with the highest expression. The median grouping strategy is used by default.

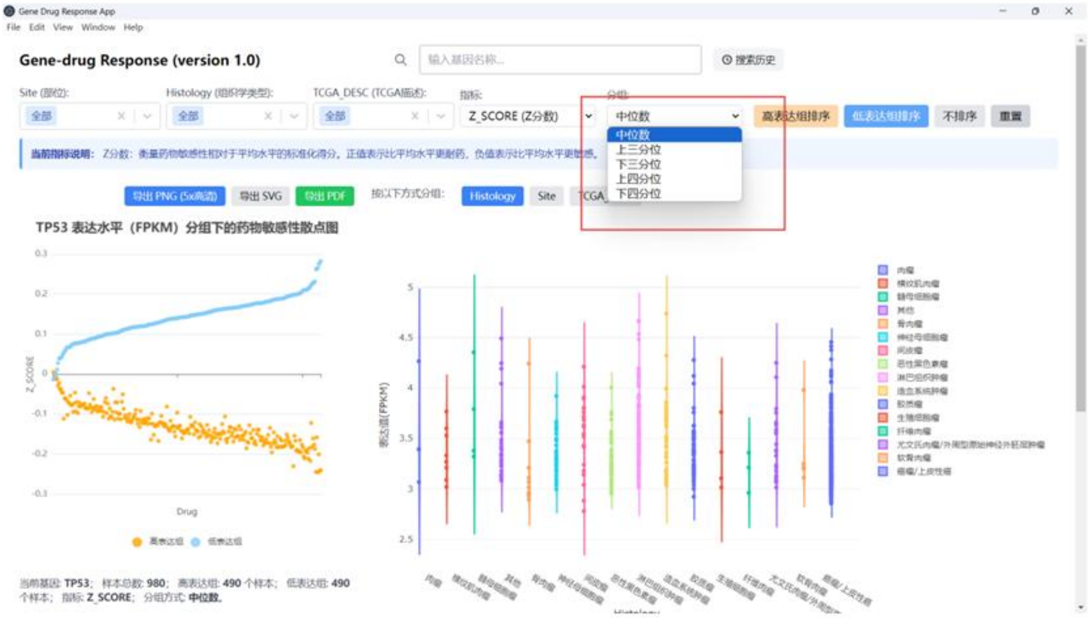

Figure 3.8 Grouping Method Selection Bar

4 Specific Operation Examples

4.1 Gene Query

(Using the BCL-2 gene as an example) Enter "BCL2" in the search bar on the left. A matching list appears in real time; click "BCL2" (Figure 4.1). The system then requests expression data for this gene from the server. After a few seconds, two plots appear in the lower area: a drug-sensitivity scatter plot grouped by BCL2 expression level (red box in Figure 4.2) and a violin plot of gene-expression values in different tissues (red box in Figure 4.3). The indicator (Z-score) and grouping method (median) are the default values at this time. (In Figure 4.2, the drug-sensitivity Z-scores of the BCL2 high-expression group are generally lower, indicating greater drug sensitivity; the Z-scores of the low-expression group are generally higher, indicating greater drug resistance.)

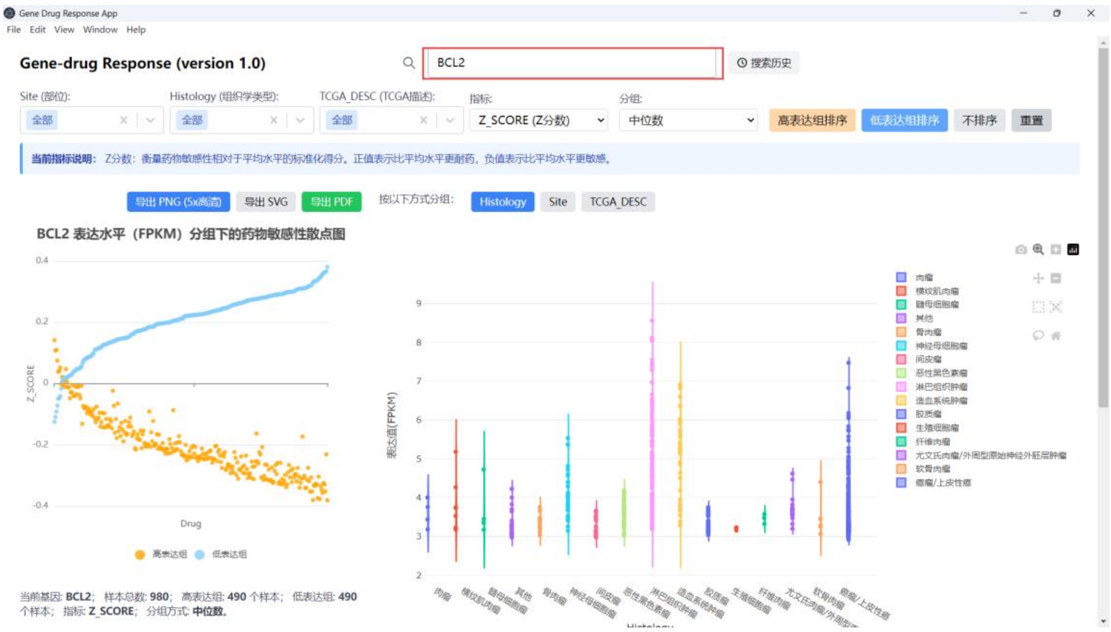

Figure 4.1 Entering BCL2 in the Gene Selection Bar

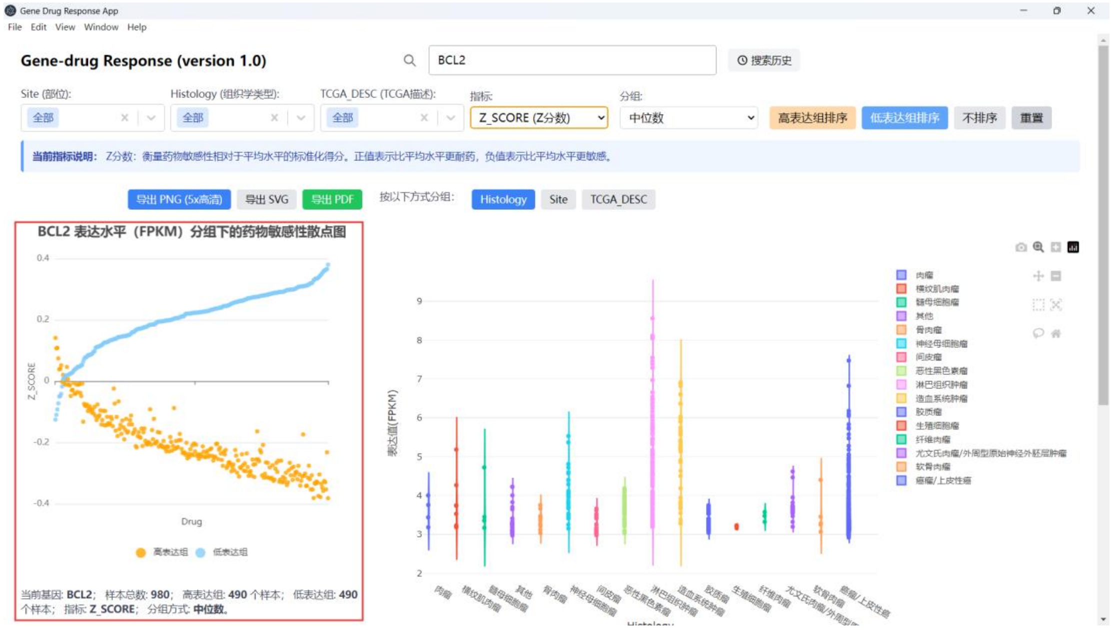

Figure 4.2 Drug-Sensitivity Scatter Plot Grouped by BCL2 Expression Level

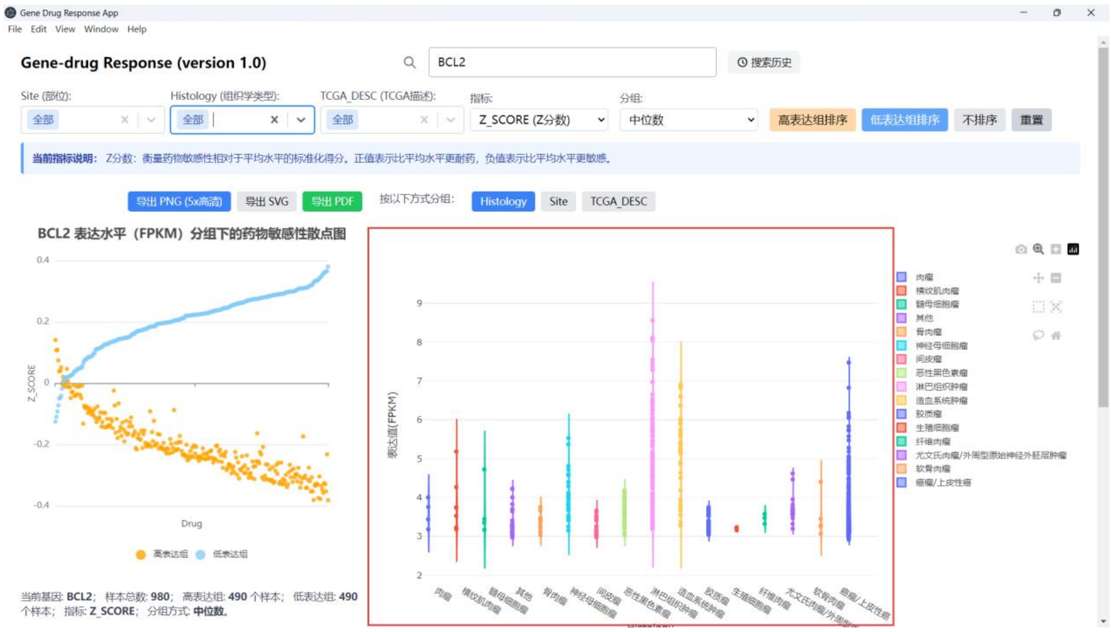

Figure 4.3 Violin Plot of Gene-Expression Values in Different Tissues

4.2 Setting the Histological Type

By default, a violin plot displays the distribution of BCL2 expression values across tissue types, with different colors representing different tumor origins (such as lung, blood, and liver). Users can select "hematopoietic system tumor" from the tissue-filter drop-down box (red box in Figure 4.4). The interface then updates the drug-sensitivity scatter plot grouped by BCL2 expression level in hematopoietic system tumors (red box in Figure 4.5) and displays cell-line data of blood origin (only violin plots related to the hematopoietic system remain in the red box in Figure 4.6).

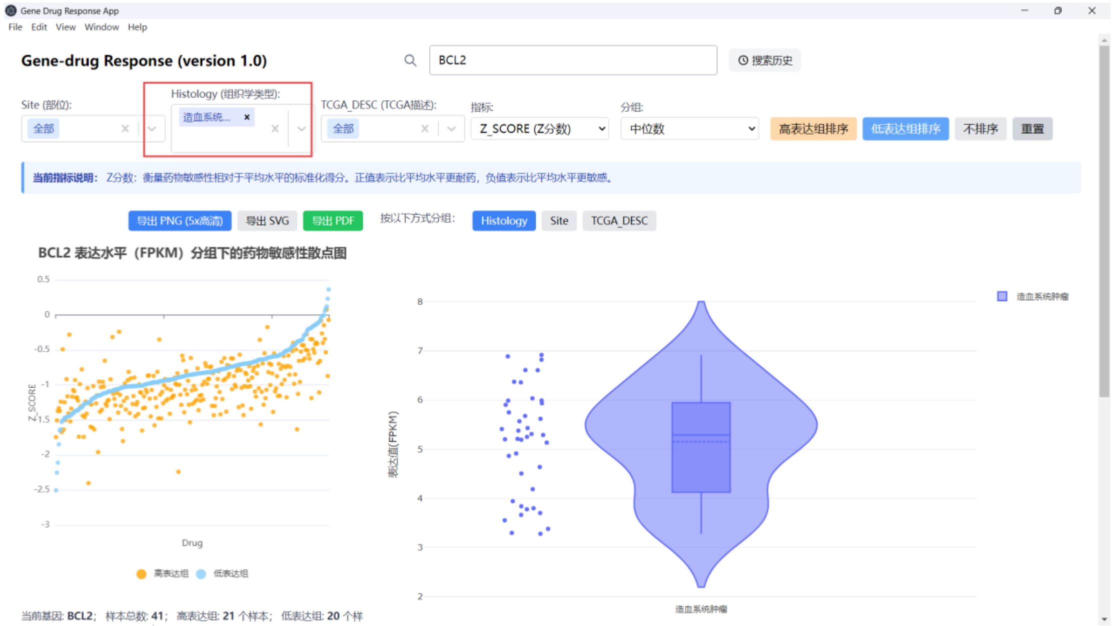

Figure 4.4 Histological Type Set to Hematopoietic System Tumor

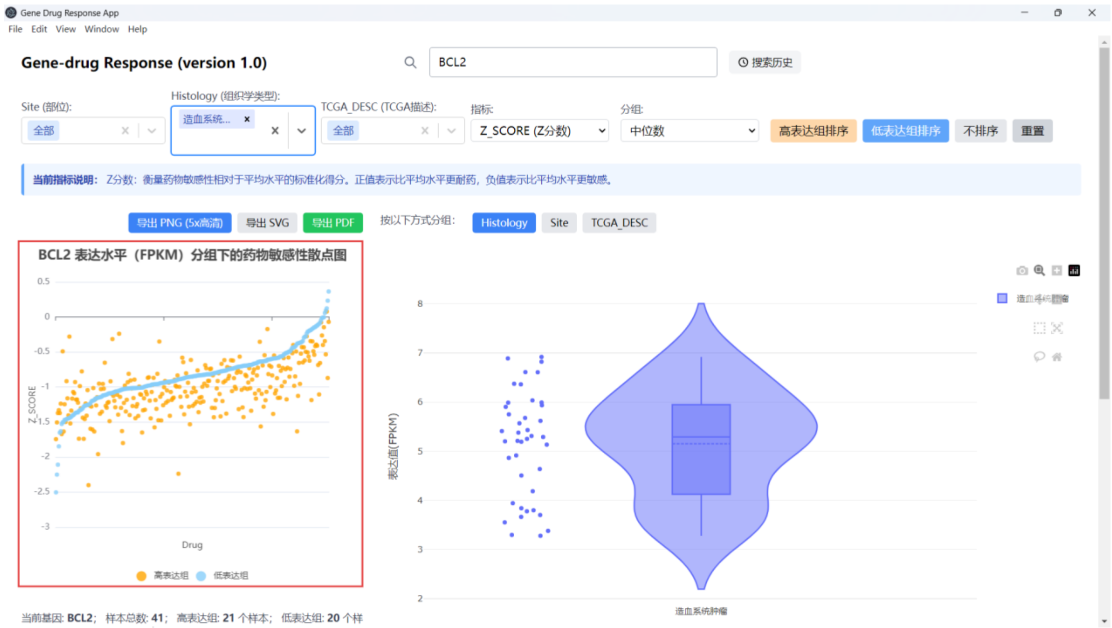

Figure 4.5 Drug-Sensitivity Scatter Plot Grouped by BCL2 Expression Level in Hematopoietic System Tumors

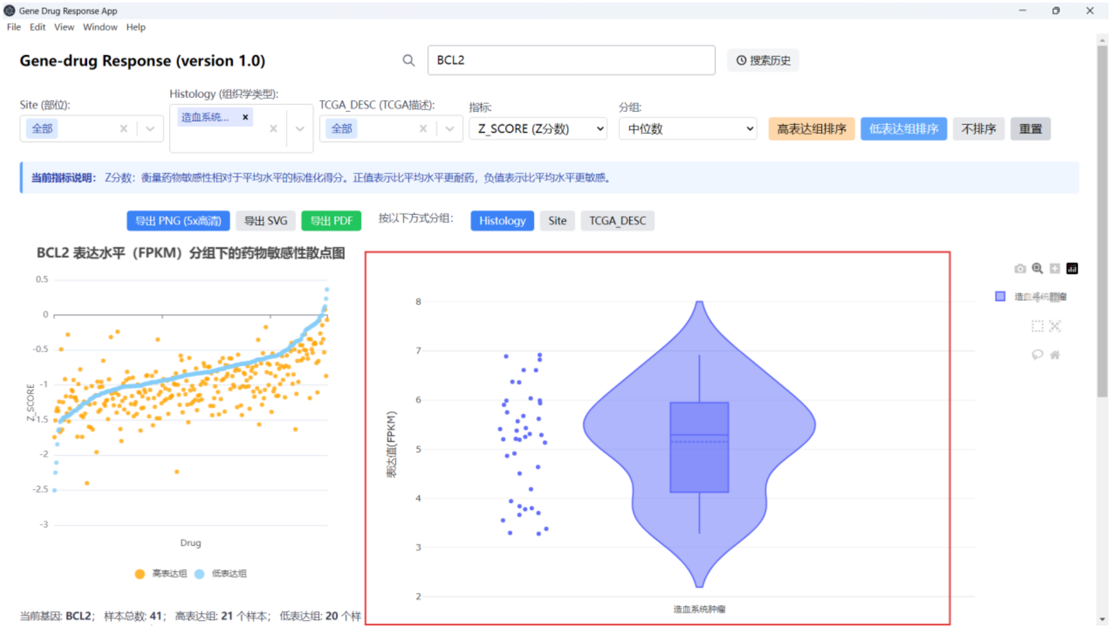

Figure 4.6 Violin Plot Related to Hematopoietic System Cell-Line Data

4.3 Setting the Grouping

Select "upper quartile" under "Grouping Method" (red box in Figure 4.7). Based on BCL2 expression values, the platform assigns the top 25% of samples among hematologic tumor cell lines to the high-expression group and the remaining 75% to the low-expression group (the remaining middle 50% of samples are temporarily excluded from in the high-low comparison). This operation completes the grouping calculation in the background and prepares the corresponding lists of cell-line IDs for subsequent analysis.

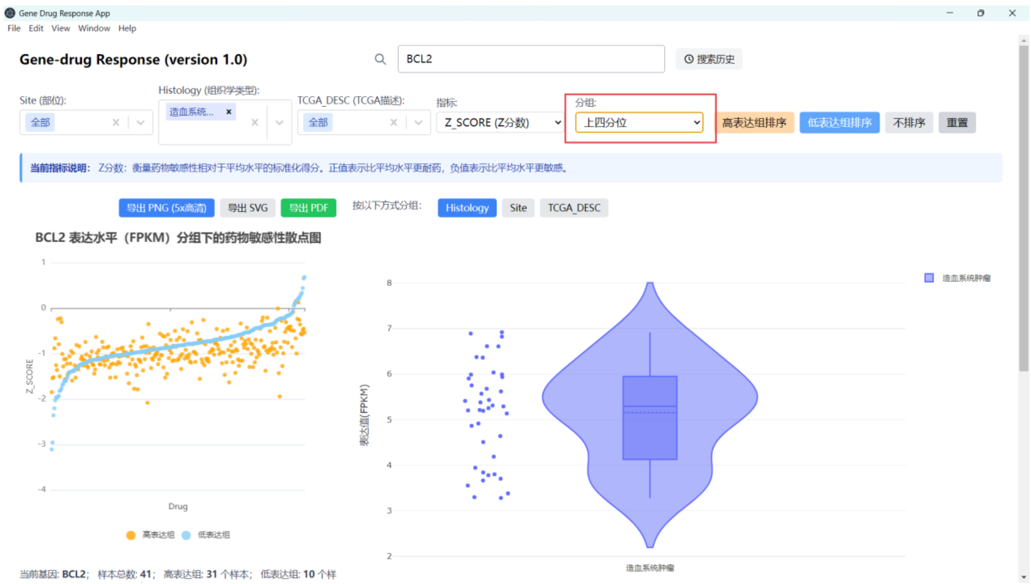

Figure 4.7 Expression-Value Grouping Method Set to Quartile

4.4 Selecting a Drug Indicator

Z-score is generally selected by default. For demonstration purposes, IC50 is selected here. Select "LN_IC50" under "Indicator Selection" (red box in Figure 4.8). The system then switches the drug-sensitivity measure to the natural logarithm of IC50 to analyze the strength of the drug effect (a smaller value indicates greater sensitivity). The indicator explanation is displayed below the interface: "LN_IC50 is the natural logarithm of the concentration that inhibits 50% of cell growth; the smaller the value, the better the drug effect."

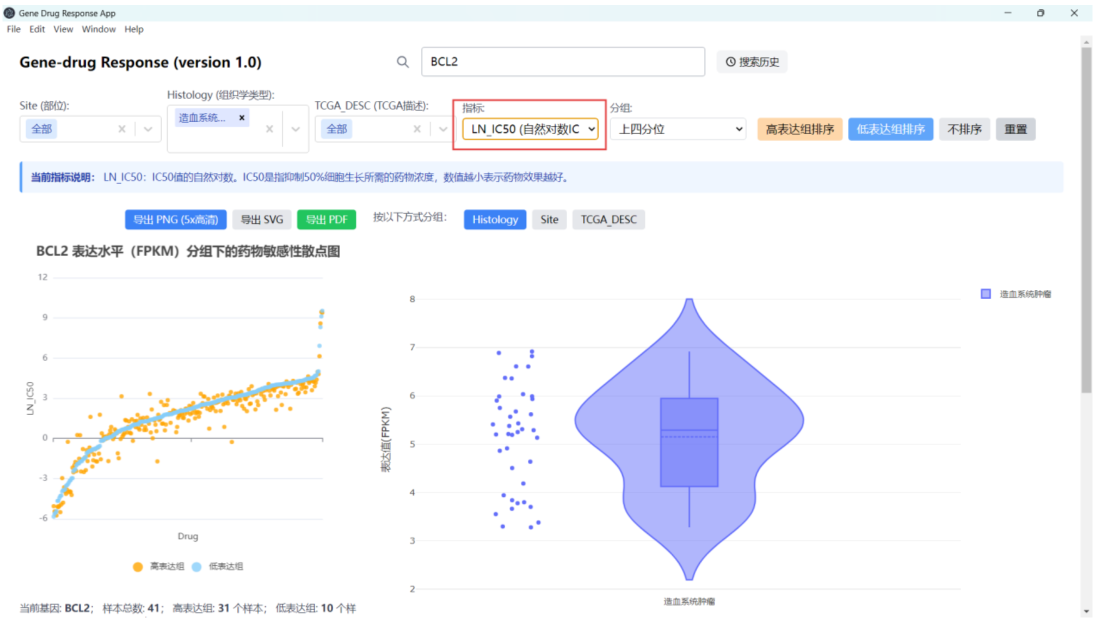

Figure 4.8 IC50 Indicator Selection and IC50 Description

4.5 Drug Selection and Analysis

4.5.1 Drug Selection

By selecting a point in the scatter plot, users can view drugs related to BCL2 or all drugs, including, for example, "Venetoclax (BCL2 inhibitor)" and "ABT-737 (BCL2/BCL-XL inhibitor)." Select "ABT-737" (red box in Figure 4.9). The system simultaneously sends two requests to the server to obtain response data for the high-expression and low-expression cell lines under this drug condition.

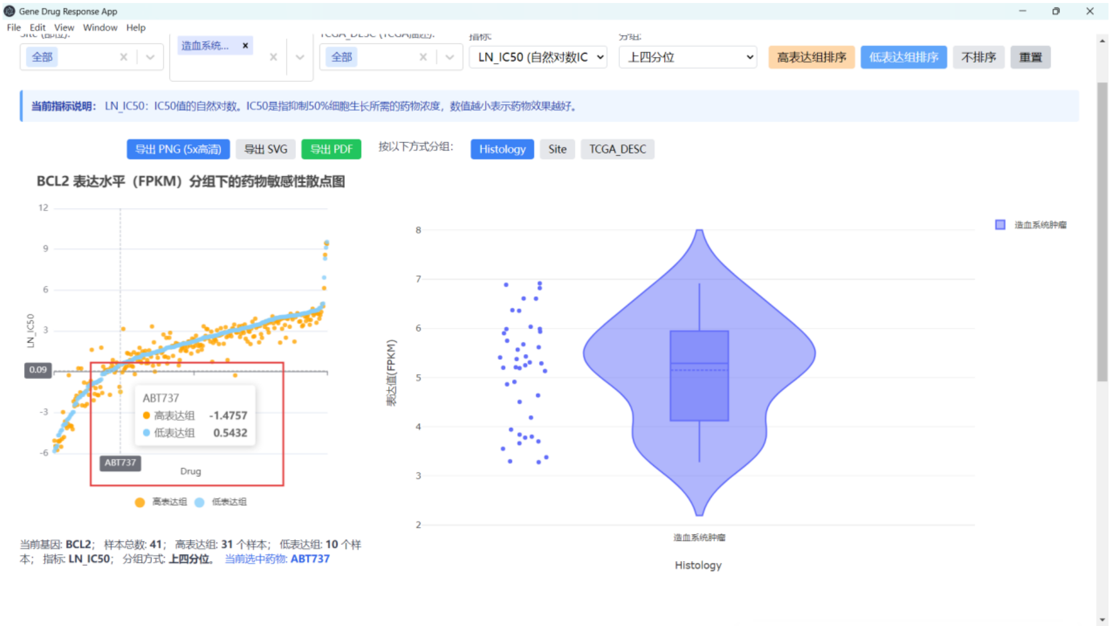

Figure 4.9 Selecting ABT737

4.5.2 Drug Analysis

After the data are received, the lower interface adds a violin plot of the IC50 distribution of the drug in the high- and low-expression groups (Figure 4.10) and a scatter plot of the correlation between gene-expression values and drug IC50 (Figure 4.11), jointly displaying the distribution differences between the high- and low-expression groups for the selected indicator (LN_IC50).

In Figure 4.10, the BCL2 high-expression cell lines can be observed to be more sensitive to ABT-737 (lower IC50), which is reflected by the lower distribution position of the violin plot for the high-expression group.

In Figure 4.11, the correlation coefficient between BCL2 gene-expression level and sensitivity to ABT-737 is r = -0.449, indicating a negative correlation between the two. In other words, the higher the BCL2 gene expression, the lower the IC50 value of ABT-737 and the better the drug effect.

At the same time, the drug-details table is displayed on the right (Figure 4.12), listing the name of ABT-737, its target (BCL2), and its pathway of action (Apoptosis_regulation, apoptosis regulation), helping users understand the drug mechanism.

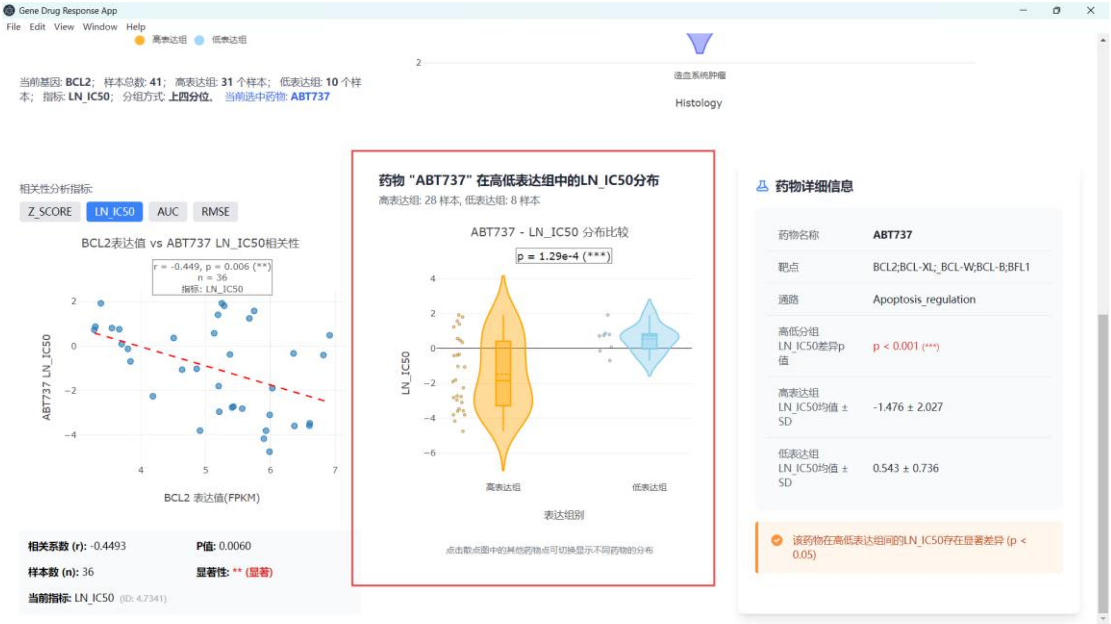

Figure 4.10 Violin Plot of the IC50 Distribution of the Drug in the High- and Low-Expression Groups

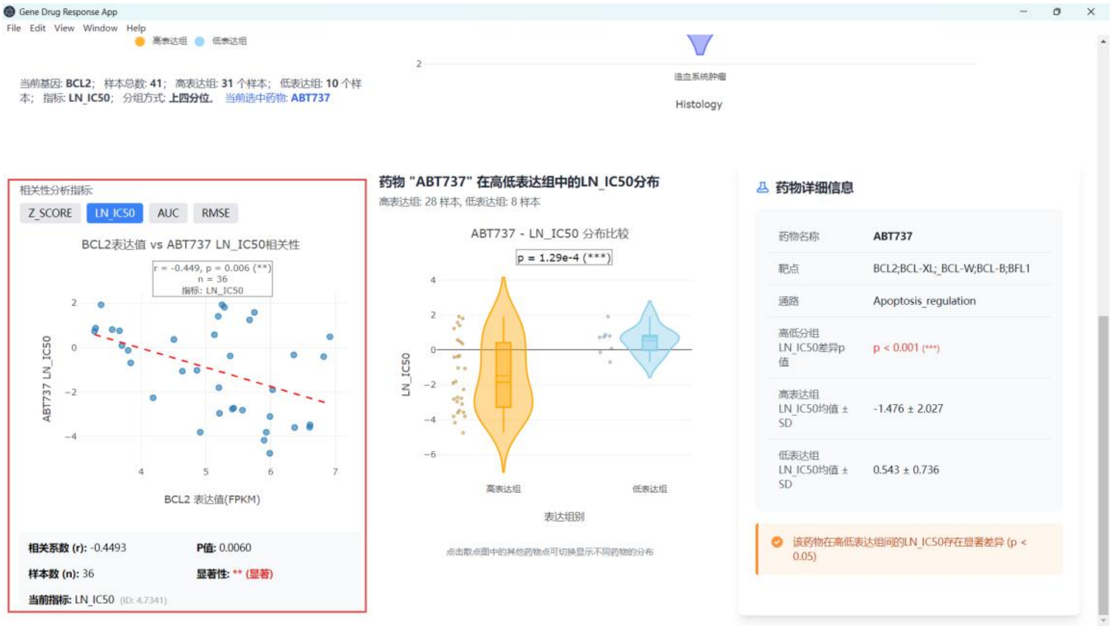

Figure 4.11 Scatter Plot of the Correlation Between BCL2 Gene-Expression Values and Drug IC50

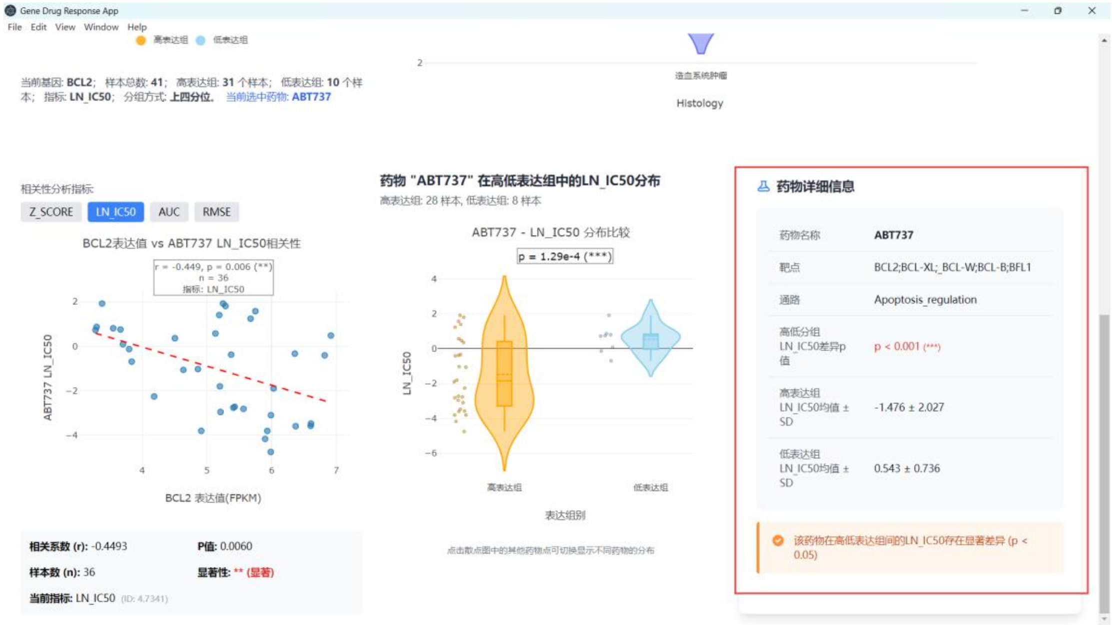

Figure 4.12 Details of Drug ABT-737

4.6 Export and Report

To save analysis results, users can click the export button in the chart area and select PDF or SVG format to download the current image (red box in Figure 4.13). In addition, the software supports exporting analysis data in PNG format for further research. After completing a gene-drug analysis, users can directly switch to another gene or drug and repeat the above steps. Note that each time the gene is changed, the previously selected drug is reset and must be selected again to refresh the violin plot.

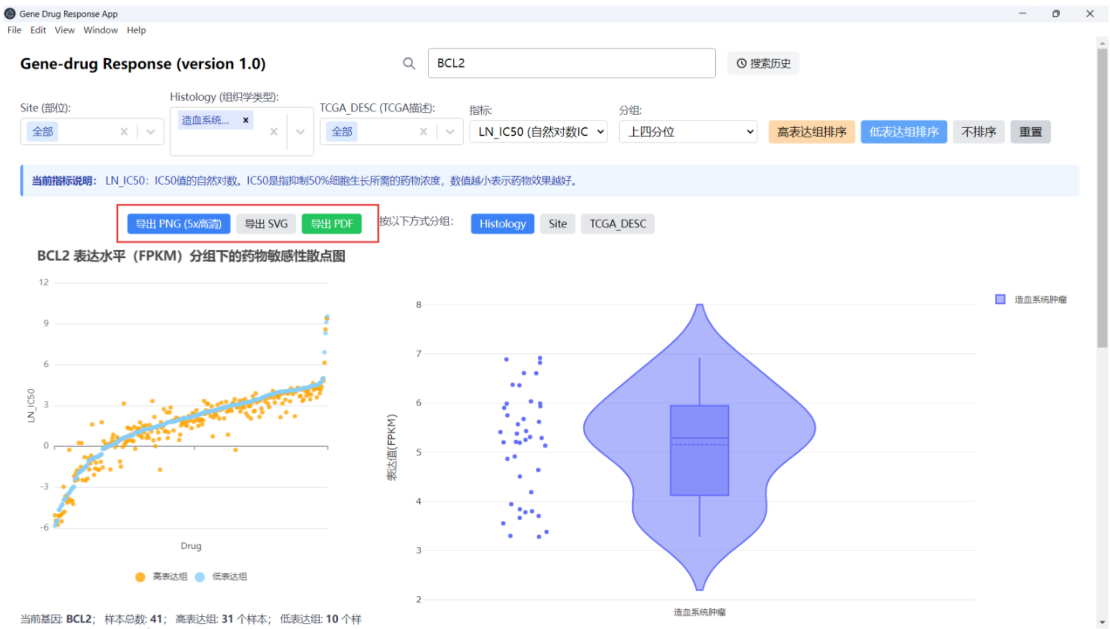

Figure 4.13 SVG, PDF, and PNG Export Options

4.7 Notes

1. During use, note that the entered gene name must strictly match the gene identifier in the dataset (the system automatically handles case sensitivity).

2. When certain tissue types are filtered, if the number of samples in the category is too small, the statistical results may not be representative, and the interface will also indicate that the sample size is insufficient.

3. If the software fails to load when opened for the first time, first click the start-backend.bat file in the folder (red box in Figure 4.14), and then click gene-drug-app.exe to open the program. Subsequently, the software can be opened directly.

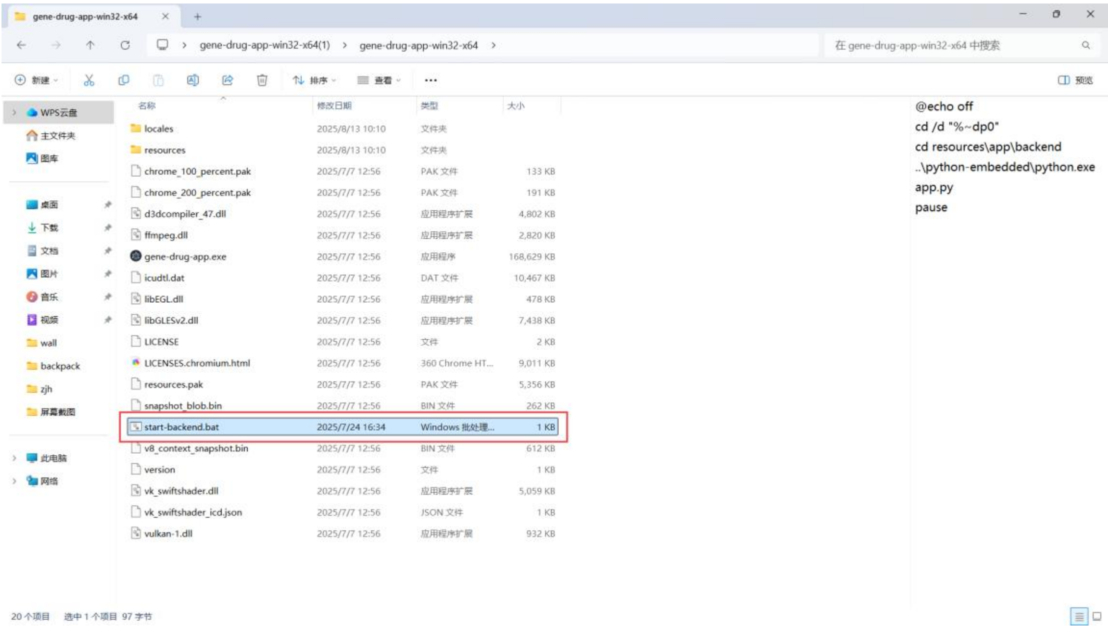

Figure 4.14 Solution When the Software Fails to Load